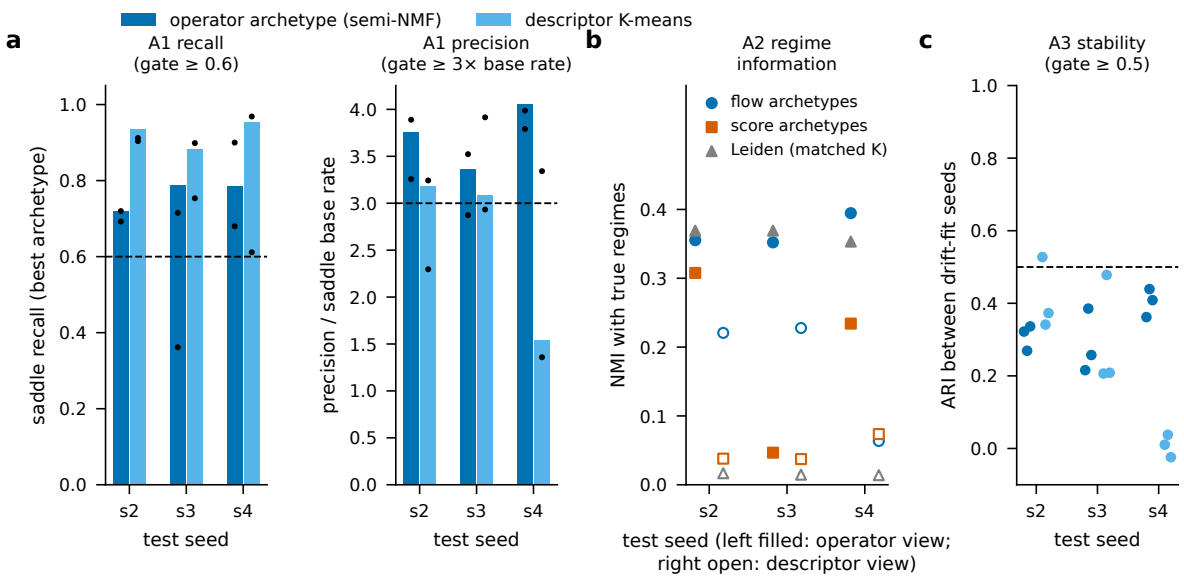
operator archetype (semi-NMF)
descriptor K-means
a
A1 recall
(gate ≥ 0.6)
saddle recall (best archetype)
1.0
0.8
0.6
0.4
0.2
0.0
s2
s3
s4
test seed
A1 precision
(gate ≥ 3× base rate)
precision / saddle base rate
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
s2
s3
s4
test seed
b
A2 regime
information
flow archetypes
score archetypes
Leiden (matched K)
NMI with true regimes
0.4
0.3
0.2
0.1
0.0
s2
s3
s4
test seed (left filled: operator view;
right open: descriptor view)
c
A3 stability
(gate ≥ 0.5)
ARI between drift-fit seeds
1.0
0.8
0.6
0.4
0.2
0.0
s2
s3
s4
test seed